Sathmayi Sridhar

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA **December 2027**
Master of Science in Quantitative Biology and Bioinformatics
Relevant Courses: Programming for Scientists, Essential Math and Statistics for Scientists, Practical Computing for Biologists

Vellore Institute of Technology, Vellore, India **June 2026**
Bachelor of Technology in Biotechnology - CGPA: 3.9/4.0

SKILLS

Languages & Computing: Python, R, Java, SQL, MATLAB, Linux/Bash, Git

Bioinformatics: RNA-seq (DESeq2), DEG, Bioconductor, Biopython, GO/KEGG enrichment, BLAST, PyMOL, Chimera, PyRx

Data Science & Statistics: Scikit-learn, ODE Modeling, SAS, JMP, Power BI, Tableau, Prism GraphPad

Certifications: IBM Introduction to Data Science, JHU Introduction to Genomic Technology, Python for Genomic Data Science

WORK EXPERIENCE

Research Intern, Systems and Computational Biology Lab – *University of Hyderabad, India* **January 2026 – June 2026**

- Engineered ODE-based mathematical models in Python (SciPy odeint) and MATLAB, incorporating physiological and pharmacological perturbations to analyze phase responses to meal administration
- Fit model parameters to experimental data and simulated circadian, inflammatory, and stress variable dynamics to identify ideal drug dosing schedules for chronotherapeutics
- Visualized data using R Studio and MATLAB for clear translation of findings to an academic panel

Research Intern, Molecular Parasitology Lab – *IBAB, Bangalore, India* **July 2025 – November 2025**

- Analyzed *P. falciparum* RNA-seq data using DESeq2, integrating Chimera and PyMOL to evaluate Cas13 protein structure
- Optimized recombinant Cas13 expression via IPTG induction, SDS-PAGE, Western blotting, CBB staining, and affinity chromatography to maximize yield for further testing of cleavage ability in mammalian cells
- Engineered Cas13 expression vectors and transformed *E. coli* BL21 cells, using GFP-targeting guide RNAs to evaluate Cas13-mediated RNA knockdown efficiency; presented findings in weekly lab meetings

Quality Assurance and Control Intern – *OneSource Specialty Pharmaceuticals, Bangalore, India* **June 2025 – July 2025**

- Applied GLP/GDP compliance for a regulated environment, performed sampling, microbial testing, and monitoring tests and coordinated with leadership teams to maintain accurate document control and quality assessment

RESEARCH EXPERIENCE

Bioinformatics and Molecular Biology Undergraduate Research, *VIT Vellore* **August 2024 – November 2025**

- Leveraged network pharmacology methods such as gene ontology, functional enrichment, and protein–protein interaction for analysis of GC-MS data of flower extracts to understand therapeutics related to acute kidney injury
- Applied molecular docking simulations using PyRx, studied compound–gene–target relationships, and investigated key metabolic pathways involved in pathological features of AKI to predict key natural leads for target discovery
- Co-authored the resulting peer-reviewed publication in *Cell Biochemistry and Biophysics* (2026)

Biostatistics Research*, VIT Vellore* **March 2025 – October 2025**

- Collected and analyzed large-scale multifactorial data in a collaborative setting to study behavioral and socioeconomic stressors of 300 women in India using data visualization (Power BI, Tableau) to study effects on cognitive health and ability
- Developed, trained, and validated an ANN model using JMP to demonstrate associations between parameters, used descriptive statistical tests (regression, ANOVA) and Prism GraphPad for data analysis, visualization, and interpretation

LEADERSHIP AND VOLUNTEERING

STEM Chapter Lead and Undergraduate Mentor – *VIT Vellore, Dept of Biotechnology* **September 2022 – June 2026**

- Curated life science review articles for social media, organized and led technical events, mentored 20+ junior students

Outreach & Program Event Coordinator – *Youth Red Cross* **September 2022 – October 2025**

- Organized volunteer events and initiatives, fostered teamwork and community service in health advocacy and education

PUBLICATIONS

Routh, S., Tatu, A., **Sridhar, S.**, Mittal, S., & Manickam, V. (2026). **Experimental Characterization Combined with Computational Network Analysis of Metabolites Associated with Chrysanthemum morifolium–derived exosome-like Nanoparticles in Predicting Potential AKI-relevant Targets.** *Cell Biochemistry and Biophysics*, 1–26. https://doi.org/10.1007/s12013-026-02008-0